ARTICLE

PRISM: a modular system for glioma progression surveillance on MRI

[PENDING: first author's full name][a] and Jessica Zhang[b]

[a]Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA; [b]Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA

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ABSTRACT

[PROPOSED, awaiting approval] Models of post-treatment glioma MRI face known challenges: most datasets lack treatment dates, centres document progression differently, cohorts are small and the hardest situations have few cases. We built PRISM, a modular system that keeps image and treatment-timeline evidence in separate pathways, measures the segmented tumour, attaches a confidence to every scan and writes a report with a local language model. On MU-Glioma-Post, the timeline pathway detected the scan at which progression was called with an AUC of 0.82 and the image pathway with 0.71. Fitted once and locked, the models reached 0.77 on LUMIERE, and single-scan images reached 0.80 on UCSF-ALPTDG, about as high as when trained there. An audit showed what each score rested on: the treatment timeline at MU, enhancing tumour volume at UCSF. Conformal sets for the combined score kept their coverage at all three centres, and reports passed a deterministic check against their records.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

After surgery and chemoradiotherapy, patients with glioma are followed with MRI every few months, and at each scan the team must decide whether the tumour has progressed. The decision is hardest in the months after radiotherapy, when treatment can increase contrast enhancement without tumour growth. This pseudoprogression is most common in the first weeks after chemoradiotherapy and more frequent when the MGMT promoter is methylated (Taal et al. 2008; Brandes et al. 2008). In routine care, progression is therefore documented from the images together with symptoms and the course of treatment, and the date on which it is recorded depends on when the patient was scanned. A model trained on such records can learn from any of these sources.

Models for this task meet five challenges that come from the data rather than from any one method. First, most public surveillance datasets hold images but few treatment dates, so a model that needs the treatment timeline cannot be applied to them. Second, centres document progression in different ways, from dated clinical decisions to expert ratings of each scan, so a model can

CONTACT [PENDING: first author. Email: ...; ORCID: ...]

score well at two centres for different reasons. Third, cohorts are small, and at this size the choice of one test split can move an AUC by more than a change of model (An et al. 2021; Varoquaux 2018). Fourth, the situations in which the decision is hardest, early after radiotherapy or when enhancement grows, hold few cases in any one cohort, so accuracy there is rarely measured. Fifth, tumour volumes from automatic segmentation are noisy, the clinician needs to see what was measured, and patient records often cannot leave the hospital. External tests address the second and third challenges, and they are uncommon: 3 of 20 studies of automated response assessment in a recent systematic review used external test data (Shi et al. 2025), and deep learning models in radiology usually lose some accuracy on external data (Yu et al. 2022).

We asked whether a system could give the clinician, at each post-treatment scan, the image evidence and the treatment-timeline evidence separately, carry to other centres without refitting, and say how far each can be trusted there. We also wanted it to explain each score in a short report written inside the hospital.

PRISM is a modular system built for this, and we tested it on one development cohort and two external cohorts with models fitted once and plans written in advance. It scores each scan through two pathways, one from clinical data and the treatment timeline and one from measurements of the segmented tumour. It shows a combined score beside them, attaches a confidence made of a conformal prediction set, a statement of whether the pathways agree and a reliability card, and writes a report with a local language model whose every sentence is checked against the scan's record. On MU-Glioma-Post, both pathways detected the scan at which progression was called, and models fitted there held at LUMIERE and UCSF-ALPTDG. An audit of the scores showed that they rested mainly on the patient's position in treatment at MU-Glioma-Post and on the enhancing tumour volume at UCSF-ALPTDG. The conformal sets for the combined score kept their stated coverage at all three centres, and the reports were faithful to their records.

2. Methods

2.1. PRISM

PRISM handles each scan in five modules, each answering one of the challenges above (Figure 1). The timeline pathway and the image pathway keep the two kinds of evidence apart, so that the image pathway can run where no treatment dates exist and the audit can show which evidence a score used. Segmentation supplies the measurements and the images the clinician sees. The confidence addresses the rare situations, and the report presents all of it without the record leaving the hospital. The third challenge, small cohorts, is handled by how PRISM is evaluated (Section 2.3).

The timeline pathway scores a scan from clinical data and the patient's position in treatment. On MU-Glioma-Post its inputs are age, sex, IDH mutation and MGMT promoter methylation, the days since diagnosis and since surgery, the start, end and dose of radiotherapy, chemotherapy and the visit index, each taken as of the scan day. Where a cohort records no treatment dates, PRISM uses a reduced timeline that keeps the clinical inputs, the days since surgery and a flag for the first weeks after radiotherapy. At LUMIERE the flag is the rater's note that the scan was less than three months after radiotherapy; at UCSF-ALPTDG it is approximated from the start of radiotherapy. Both pathways are ridge logistic regressions, with every fitted step inside the training data.

The image pathway scores a scan from simple measurements of the segmented tumour. On

MU-Glioma-Post it uses 27 inputs: the log volumes of six tumour compartments, including enhancing tumour, necrosis, the surrounding FLAIR hyperintensity and the resection cavity, with shape and relative intensity features; the log ratio of each volume to the previous scan; and the ratio of the enhancing and total lesion volumes to the smallest volume in the patient's history. It holds no dates, intervals or treatment flags. A single-scan version keeps only the 19 measurements of the current scan. At LUMIERE, whose masks have three labels, the pathway uses three volumes and their change from the previous scan, and at UCSF-ALPTDG the history is the one earlier scan in the dataset. We defined these inputs after finding that the planned imaging model contained inputs that encode the scan schedule (Section 3.3), so the image pathway on MU-Glioma-Post is exploratory; its use at UCSF-ALPTDG and in the system tests was planned. Where both pathways are available, the combined score is one ridge model with both sets of inputs; at LUMIERE and UCSF-ALPTDG it adds harmonised volumes, their change and the interval since the previous scan to the reduced timeline.

Segmentation supplies these measurements and the images shown in the report. The measurements come from the masks distributed with each cohort: the automatic masks released with MU-Glioma-Post (Mahmoud et al. 2025), the DeepBraTumIA masks of LUMIERE (Suter et al. 2022) and the masks provided with UCSF-ALPTDG (Fields et al. 2024). From the same images and masks, PRISM's segmentation module draws the figure in each report and the volumes printed beside it. The figure is rendered by code; the language model never sees an image.

Each scan's confidence has three parts: a conformal set, a pathway-agreement statement and a reliability card. The conformal set is a patient-level split conformal prediction set at the 80% level (Angelopoulos and Bates 2023). It holds one outcome when the score is clear and both outcomes when it is not, and it should contain the true outcome for at least 80% of scans. On MU-Glioma-Post the set was calibrated inside each training fold on 30% of its patients; for the external cohorts one set, calibrated on MU-Glioma-Post, was applied unchanged. The two pathways agree when both scores lie on the same side of their median in the cohort. The reliability card looks up the scan's situation, its phase of care crossed with the change in enhancing tumour since the previous scan, in a table of results measured in each cohort, in the spirit of a model card (Mitchell et al. 2019). It states the situation's tier: tested, with AUCs and intervals, when the situation holds at least 10 progression and 10 non-progression scans; few cases, with counts only, when it holds fewer; and not tested when it holds none. The agreement rule and the card were defined after the cohort results had been seen and are exploratory.

A local language model writes the report from the scan's record, and a deterministic checker tests every sentence against that record. The model (Qwen3.8-27B) runs offline at temperature 0 and receives only the record; the checker rejects any number, flag or confidence statement that is not in the record, any omission of required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

2.2. Cohorts and the task

PRISM was developed on MU-Glioma-Post and tested, locked, on LUMIERE and UCSF-ALPTDG (Table 1). MU-Glioma-Post records the dates of surgery, radiotherapy and chemotherapy, and progression as a documented date (Mahmoud et al. 2025). LUMIERE records no treatment dates; each scan carries an expert rating of response, and the rater notes whether the scan falls within three

months of radiotherapy (Suter et al. 2022). UCSF-ALPTDG holds one pair of consecutive post-treatment scans per patient, with the date of progression and the start of radiotherapy (Fields et al. 2024). Because the first scan of each pair has no earlier scan, we evaluated the second scan only. UCSF-ALPTDG was obtained under a data use agreement. The primary population in every cohort is grade 4 glioma.

The task is detection at the scan: is this the scan at which progression is called? On MU-Glioma-Post a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. At UCSF-ALPTDG a positive is a second scan with progression dated within 30 days of it and nearer to it than to the first scan; a negative is a second scan with progression at least 120 days later, or with at least 120 days of follow-up without progression. On MU-Glioma-Post the intended use is scans from the start of radiotherapy onwards, and every MU-Glioma-Post result below is for this population. We divide it into phases of care: during radiotherapy, early after radiotherapy (up to 84 days after its end) and later. Predicting progression within 120 days of a scan, the task with which this work began, is reported as a secondary task.

2.3. Evaluation

On MU-Glioma-Post, every model was evaluated by repeated patient-grouped cross-validation, because a single split moved scores more than any model choice (Supplementary Section S8). We used five repeats of five folds, with all scans of a patient in the same fold, and report the mean over repeats of the pooled held-out AUC with a 95% confidence interval (CI) from 1,000 patient-level bootstrap draws. The need for this design showed in our own first evaluation: the initial radiomics model reached an AUC of 0.80 on one test split, and when it was re-selected and re-tested over 500 splits of the same design it averaged 0.72. At UCSF-ALPTDG, the dataset's own training and test split gave a timeline model trained there an AUC of 0.45, against a median of 0.72 over 100 random splits.

For the external tests, each model was fitted once on MU-Glioma-Post and applied once, under plans committed before any external label was read. The locked models were fitted on all grade 4 detection scans of MU-Glioma-Post, progression and non-progression scans from every phase of care, and external AUCs carry 95% CIs from 1,000 patient-level bootstrap draws. As a reference, we also fitted each UCSF-ALPTDG model on UCSF-ALPTDG itself by repeated cross-validation. Table 2 lists the questions fixed in advance and their answers.

Analyses that were not planned are labelled exploratory wherever they appear. They are the definition of the image pathway on MU-Glioma-Post, the three-step comparison of image models, the agreement rule and the reliability card, a reduced confidence design, the clinical-only model at LUMIERE, and the second evaluation of the reports, which followed a failure seen in the first. Comparisons within a phase of care or within an enhancement category are diagnostics: they test what a score rests on and carry no decision rule.

3. Results

3.1. Each pathway carries evidence on MU-Glioma-Post

In the intended use on MU-Glioma-Post, the timeline pathway detected the scan at which progression was called with an AUC of 0.82 (95% CI 0.75–0.90), and the image pathway with 0.71 (0.63–0.78), over 146 scans of 80 patients (62 progression, 84 non-progression). Clinical data alone, without the timeline, scored 0.53, close to chance. The combined score also reached 0.82, so on top of the timeline the images added nothing measurable at MU-Glioma-Post. The planned test agreed: the planned imaging model did not add the pre-specified 0.03 to the timeline (Table 2). The secondary task was harder; predicting progression within 120 days, PRISM's model reached 0.60 (0.52–0.69) in the intended use (Supplementary Section S23). A timeline pathway needs treatment dates, which the two external cohorts largely lack, so the next question was how each pathway carries to them.

3.2. The pathways hold at two external centres

Fitted once on MU-Glioma-Post and applied unchanged, the combined model reached 0.77 (0.68–0.86) on LUMIERE, and single-scan image measurements reached 0.80 (0.69–0.89) on UCSF-ALPTDG (Figure 2). The same LUMIERE model scored 0.80 in cross-validation on MU-Glioma-Post, so it carried to LUMIERE with little loss. At LUMIERE the two kinds of evidence were complementary: images alone scored 0.65 and the reduced timeline alone 0.69, both below the combined score. Clinical data alone stayed close to chance in all three cohorts (0.55 at LUMIERE, exploratory, and 0.50 at UCSF-ALPTDG).

At UCSF-ALPTDG, the locked single-scan image model was about as high as when trained there (0.83). For the image pathway with change features, training at UCSF-ALPTDG did better: the locked model scored 0.73 and the UCSF-trained model 0.86 (difference 0.14, 0.03–0.26). The locked image pathway met its planned transport rule (Table 2), but its change inputs carried less well than the single-scan measurements; change between the two UCSF scans spanned a range three to six times narrower than at MU-Glioma-Post, which suggests why. The reduced timeline transported least, at 0.56; UCSF-ALPTDG records only the start of radiotherapy, so its timeline inputs are approximate. We read these results as transport of PRISM's measurements to a new centre; they do not show that a model fitted elsewhere does better than one fitted at UCSF-ALPTDG. Similar AUCs at different centres can rest on different evidence, which the audit examines next.

3.3. What each score rests on depends on the cohort

The audit shows that on MU the scores rest mainly on where the patient is in treatment, and on UCSF on how much the tumour enhances (Figure 3). At MU-Glioma-Post the scan schedule separates the classes before any image is read: 56 of the 62 progression scans came more than 12 weeks after radiotherapy, while 51 of the 84 non-progression scans came in the 12 weeks after it. When we compared models within each phase of care, the timeline's gain over clinical data and images, 0.11 (0.04–0.20) across phases, fell to −0.02 (−0.13 to +0.11) within them (exploratory; Figure 3A). Within a phase, images alone reached 0.68 (0.58–0.77) and the timeline 0.64 (0.51–0.78), so there the images carried about as much evidence as the timeline.

At UCSF-ALPTDG the scores rest on enhancement. Enhancing tumour volume alone gave an

AUC of 0.88 (0.79–0.95), higher than the locked single-scan image model, and UCSF rated enhancement as increased at 27 of the 29 progression scans (Figure 3B). Progression was dated a median of 7 days after the evaluated scan, and time since diagnosis, which separates the classes at MU-Glioma-Post, gave 0.54 at UCSF-ALPTDG. The UCSF AUCs therefore measure agreement with a call made from enhancement at the same scan. The combined score's higher AUC at UCSF-ALPTDG (Figure 2) rests partly on the interval between the two scans, which was shorter before progression (exploratory).

Each step of the image model's design raised its AUC on MU, and the audit shows what each step added (exploratory; Figure 3C). Measurements of the current scan alone reached 0.66, and adding change from earlier scans gave the image pathway of Section 3.1. Adding the interval since the previous scan, a flag for whether there was one and the smallest earlier volume counted only from the end of radiotherapy gave 0.81, close to the timeline itself. Within a phase of care the last step added nothing (−0.01, −0.06 to +0.04), because those three inputs encode the scan schedule. They form the imaging block of the planned model, and they are the reason PRISM's image pathway stops at the second step.

Deep networks enter PRISM as segmenters, and the masks of MU-Glioma-Post and LUMIERE were made by automatic segmentation. We re-segmented MU-Glioma-Post with a SwinUNETR network trained on BraTS 2021 as a pre-specified sensitivity analysis; its enhancing tumour masks agreed with the cohort's own (median Dice 0.80), and detection from its measurements did not change (−0.02, −0.06 to +0.03; Supplementary Section S14). When we fine-tuned such networks to score UCSF scans directly, they scored above a permuted-label control (Table 2), but their scores followed enhancing volume (correlation about 0.8) and their AUC stayed below that of enhancing volume alone; on MU-Glioma-Post they did not separate the scans. The networks therefore contribute the masks, and through them the measurements and the figure in each report. Because the evidence behind a score differs by cohort and by situation, PRISM also has to say how far each score can be trusted.

3.4. The confidence keeps its coverage at all three centres

At the 80% level, the conformal sets of the combined score covered 0.82 of MU scans, 0.87 of LUMIERE scans and 0.78 of UCSF scans, against a threshold of 0.78 fixed in advance (Figure 4A). The point estimate met the threshold in all three cohorts, although the UCSF interval reached below it (0.68–0.87). Most sets were informative, holding a single outcome for 62% to 75% of scans. Coverage fell short for the image pathway alone at LUMIERE and the timeline alone at UCSF-ALPTDG, and it was uneven across situations within each cohort ([Supplementary section for the system results, to be added]).

When the two pathways disagreed on MU, the combined score was less accurate, and setting those scans aside raised the AUC of the rest more than removing the same number of scans matched for phase of care. Disagreement marked 40% of MU scans. The combined score's AUC was 0.90 where the pathways agreed and 0.59 where they disagreed (difference 0.31, 0.18–0.43), and setting the disagreeing scans aside gained 0.08 (0.05–0.12) over matched removal (Figure 4B). At LUMIERE and UCSF-ALPTDG disagreement did not pick out the less accurate scans (Table 2). We then tried a reduced design that flags a scan only when the pathways disagree and its conformal set holds both outcomes (exploratory, chosen after these results were read). It flagged 12% to 27% of scans, and

flagged scans were wrong about half the time on MU-Glioma-Post and LUMIERE, against about a fifth for the rest; at UCSF-ALPTDG the difference was smaller and its interval included zero. This design has not been tested on new data.

For most scans the evidence behind any score is thin, and the reliability card says so (exploratory). Of the 249 MU scans with reports, 35 fell in a situation tested on at least 10 progression and 10 non-progression scans. Across the three cohorts only four situations met that bar, one at MU-Glioma-Post and three at LUMIERE, and every UCSF situation had few cases (Figure 4C). At MU-Glioma-Post no situation in the first 12 weeks after radiotherapy held more than three progression scans. The tier states how much evidence lies behind a situation's AUC. It tracked error at LUMIERE and ran the other way at MU-Glioma-Post, where the scans with few cases were mostly non-progression scans early after radiotherapy, which are easy to score. The report is where the clinician meets these statements.

3.5. The reports are faithful to the record

Every evaluation report written by the local model passed the deterministic check against its record: 114 of 114 on MU, 162 of 162 on LUMIERE and 130 of 130 on UCSF (Figure 5). These counts come from a second evaluation, on patients not used in development, after the first had failed at LUMIERE (0 of 178) because one sentence quoted a number that was in no cited field; we corrected the prompt and left the checker unchanged. The reports stated the evidence tier for all 220 scans whose evidence was thin and followed a changed field in 28 of 30 perturbed records, so the clinician reads the segmentation, both pathway scores and the confidence as they stand in the record. Of seven open-weight models (descriptive), only Qwen3.8-27B and Gemma-4-31B passed in every cohort; smaller models mostly left required content out, and the 2B model also invented numbers.

4. Discussion

PRISM gives the clinician separate image and treatment-timeline evidence at each scan, carries to two external centres without refitting, and states how far its scores can be trusted there. At the development centre the timeline carried most of the evidence, largely because of when patients were scanned, and within a phase of care the images carried about as much. At UCSF-ALPTDG, where few treatment dates are recorded, the image measurements did the work, and their scores matched the enhancement-based call made at the same scan. Because the pathways were kept apart, these differences could be measured at each centre, and the confidence and the card carry them into each report.

Because its modules are separate, PRISM runs where data are incomplete. The image pathway needs only the scans and their masks, which is what most public surveillance datasets hold; at LUMIERE and UCSF-ALPTDG it ran without treatment dates. Where dates exist, the timeline is added as a second pathway and the combined score as a display beside both, without changing the image model. A new cohort adds rows to the reliability card, so the tiers change as evidence accumulates and no confidence model has to be refitted.

Consider a scan eight weeks after radiotherapy in which the enhancing tumour has grown. At MU-Glioma-Post, 1 of the 9 scans in this situation was progression. The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image

pathway scores it high, because enhancement has grown. PRISM's report would show both scores, state that they disagree, give a conformal set that holds both outcomes and record that the card has few cases for this situation. The clinician then sees that the system has no measured basis for a call at this scan, in the weeks when pseudoprogression is most common, and decides with information PRISM does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each of PRISM's components has been described before, but we found no system that combines them. Automatic segmentation of post-treatment glioma with measurement of change between scans is established (Rudie et al. 2022), conformal prediction has been applied to glioma segmentation (Elyassirad et al. 2025), clinical and imaging data have been fused to separate pseudoprogression from progression (Gomaa et al. 2025), and model cards report performance by subgroup (Mitchell et al. 2019). The closest system we found combines treatment variables extracted by a language model with tumour volumes to assign a structured follow-up category at one centre, and it reports no uncertainty (Jabal et al. 2026). On MU-Glioma-Post, a radiomics model reached an AUC of 0.80 with a label assigned at the patient level and one held-out set (Christodoulou et al. 2026), so its result and ours answer different questions. On LUMIERE, a classifier of four response categories reached a mean AUC of 0.81 [cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified], and Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres; both address other labels.

This study has limits. PRISM was developed at one centre, with 126 grade 4 patients, and the two external tests are its only evidence of transport. Progression is the progression documented by each centre, without expert re-reading of the scans, and at UCSF-ALPTDG it agrees closely with an enhancement call at the same scan, while a quarter of the non-progression scans there show no enhancement at all. Subgroups are small: UCSF-ALPTDG contributes 29 progression scans, and its intervals are 0.15 to 0.25 wide. The three-step image comparison, the agreement rule, the card and the reduced confidence design are exploratory and need testing on new data. The report checker is lexical, so it can confirm that every number and statement is in the record but cannot judge how a clinician reads the report, and no clinician has yet read the reports. PRISM's 120-day forecast was weak in the intended use, so we present it as a detection system.

The next step is to pool the rare situations across cohorts and train on them. Early progression after radiotherapy and growing enhancement without progression are rare at any one centre. The Burdenko and UCSD-PTGBM datasets add radiotherapy detail and advanced sequences, and each new cohort adds cases to the card's thin situations, which is where models fine-tuned on these long-tail cases could be tested. A reader study in which clinicians use PRISM's reports at the scan would show whether separate evidence and a stated confidence change their decisions.

Conclusion

PRISM shows that a surveillance system can keep its kinds of evidence apart, carry them to other centres and say where they have and have not been shown to work. In our three cohorts the images and the treatment timeline each carried evidence, in different amounts at different centres, and a clinician using PRISM can see which is which at every scan.

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Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight models in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

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References

An C et al. 2021. Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results. PLOS ONE. 16(8):e0256152. https://doi.org/10.1371/journal.pone.0256152

Angelopoulos AN, Bates S. 2023. Conformal prediction: A gentle introduction. Found Trends Mach Learn. 16(4):494–591. https://doi.org/10.1561/2200000101

Brandes AA, Franceschi E, Tosoni A et al. 2008. MGMT promoter methylation status can predict the incidence and outcome of pseudoprogression after concomitant radiochemotherapy in newly diagnosed glioblastoma patients. J Clin Oncol. 26(13):2192–2197. https://doi.org/10.1200/JCO.2007.14.8163

Chen C, Guo F, Huang D et al. 2026. Predicting recurrence risk of glioblastoma based on preoperative-postoperative longitudinal MRI: A multicenter study. Bioengineering. 13(6):668. https://doi.org/10.3390/bioengineering13060668

Christodoulou RC, Vamvouras G, Pitsillos R, Solomou EE, Georgiou MF. 2026. Explainable radiomics with probability calibration for postoperative glioblastoma surveillance. Eur J Radiol Artif Intell. 5:100074. https://doi.org/10.1016/j.ejrai.2026.100074

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, Faghani S. 2025. CONSeg: Voxelwise uncertainty quantification for glioma segmentation using conformal prediction. Am J Neuroradiol. 46(12):2553–2560. https://doi.org/10.3174/ajnr.A8914

Fields BKK et al. 2024. The University of California San Francisco adult longitudinal post-treatment diffuse glioma MRI dataset. Radiol Artif Intell. 6(4):e230182. https://doi.org/10.1148/ryai.230182

Gomaa A, Huang Y, Stephan P et al. 2025. A self-supervised multimodal deep learning approach to differentiate post-radiotherapy progression from pseudoprogression in glioblastoma. Sci Reports. 15:17133. https://doi.org/10.1038/s41598-025-02026-7

Jabal MS et al. 2026. Agentic automation of BT-RADS scoring: End-to-end multi-agent system for standardized brain tumor follow-up assessment. [Preprint, arXiv:2603.21494]. `https://arxiv.org/abs/2603.21494`

Mahmoud E, Gass J, Dhemesh Y et al. 2025. MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features. Sci Data. 12(1):1847. https://doi.org/10.1038/s41597-025-06011-7

Mitchell M, Wu S, Zaldivar A et al. 2019. Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). ACM. p 220–229

Rudie JD, Calabrese E, Saluja R et al. 2022. Longitudinal assessment of posttreatment diffuse glioma tissue volumes with three-dimensional convolutional neural networks. Radiol Artif Intell. 4(5):e210243. https://doi.org/10.1148/ryai.210243

Shi T, Kujawa A, Linares C, Vercauteren T, Booth TC. 2025. Automated longitudinal treatment response assessment of brain tumors: A systematic review. Neuro-Oncology. 27(8):1946–1971. https://doi.org/10.1093/neuonc/noaf037

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Taal W, Brandsma D, de Bruin HG et al. 2008. Incidence of early pseudo-progression in a cohort of

malignant glioma patients treated with chemoirradiation with temozolomide. Cancer. 113(2):405–410. https://doi.org/10.1002/cncr.23562

Varoquaux G. 2018. Cross-validation failure: Small sample sizes lead to large error bars. NeuroImage. 180:68–77. https://doi.org/10.1016/j.neuroimage.2017.06.061

Yu AC, Mohajer B, Eng J. 2022. External validation of deep learning algorithms for radiologic diagnosis: A systematic review. Radiol Artif Intell. 4(3):e210064. https://doi.org/10.1148/ryai.210064

Tables

Table 1. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross-validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 index scans (median 2)	Up to 10 index scans (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	None; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Progression documented as	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: progression / non-progression scans (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; non-progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary 120-day task at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG.

[b] Positives are the scan at which progression was called; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the 120-day cohort because it includes patients who contribute only a progression scan. At MU-Glioma-Post, 47 progression scans of patients with no earlier non-progression scan are excluded.

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six-week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 2. Questions fixed in advance and their answers. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Result	Met
Detection on MU-Glioma-Post			
Do image measurements add to the timeline?	Planned model ≥ 0.75 (lower bound ≥ 0.65) and gain ≥ 0.03 with CI above 0	Model 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)	No
Do fine-tuned deep networks add?	Gain ≥ 0.03 with CI above 0; permuted-label control covers 0.5	No network added; control 0.483 (0.436–0.527)	No (control valid)
Does the locked model transport to LUMIERE?	External AUC ≥ 0.65	Lower bound above 0.65 (Section 3.2)	Yes
UCSF-ALPTDG			
Q1. Does the image pathway transport?	AUC ≥ 0.65, lower bound > 0.5	Lower bound 0.603	Yes
Q2. Do images add to clinical data without a timeline?	Gain with CI above 0	+0.248 (+0.096 to +0.392)	Yes
Q3. Does training at UCSF-ALPTDG do better than the locked model?	Descriptive	Higher when trained there (Section 3.2)	Descriptive
Q4. Do deep networks learn at UCSF-ALPTDG?	Above the permuted-label control; control covers 0.5	+0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)	Yes
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)			
S1. Do conformal sets keep their coverage?	Coverage ≥ 0.78 at the 80% level in each cohort	Combined score met in all three (Section 3.4); images at LUMIERE 0.708 and timeline at UCSF-ALPTDG 0.696	Yes (combined)
S2. Do flags pick out less accurate scans?	AUC unflagged minus flagged > 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); LUMIERE +0.105 (−0.049 to +0.266); UCSF-ALPTDG −0.019 (−0.218 to +0.175); any flag at MU-Glioma-Post −0.170 (−0.503 to +0.147)	MU only
S3. Does setting flagged scans aside beat matched removal?	Retained AUC minus matched removal > 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); LUMIERE +0.051 (−0.028 to +0.144); UCSF-ALPTDG −0.004; set holding both outcomes met at UCSF-ALPTDG, +0.073 (+0.014 to +0.138)	MU only (disagreement)
S4. Do the card's tiers mean something?	Descriptive: tested below few cases in Brier score and error	Brier 0.187 vs 0.258 at LUMIERE; reversed at MU-Glioma-Post, 0.222 vs 0.147	Descriptive
S5. Are reports faithful to the record?	≥ 95% pass the checker in each cohort	First evaluation: LUMIERE 0 of 178; second evaluation: all passed (Section 3.5)	No, then yes
S6. Is the tier stated where evidence is thin?	100%	All (Section 3.5)	Yes
S7. Do reports follow	≥ 27 of 30 perturbed	29 of 30 in the first evaluation;	Yes

Figure legends

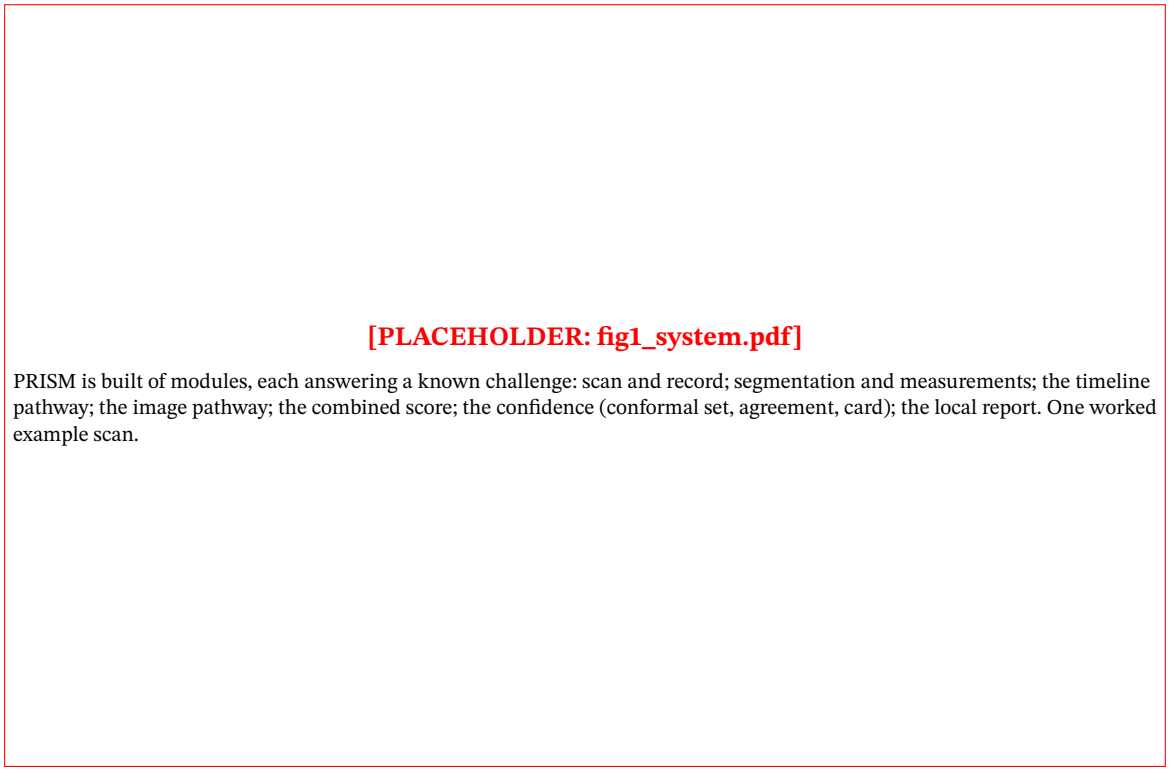


Figure 1. PRISM is built of modules, each answering a known challenge. A scan and its record enter two pathways: the timeline pathway (clinical data and the patient's position in treatment) and the image pathway (measurements from the segmentation masks). The combined score is shown beside them. The confidence has three parts: a conformal prediction set at the 80% level, a statement of whether the two pathways agree, and the reliability card's tier for the scan's situation. A local language model writes the report from the record, and a deterministic checker tests every sentence against it. One example scan is followed through each module.

[PLACEHOLDER: fig2_transport.pdf]

AUC by pathway (clinical alone, timeline, images, combined) and cohort (MU internal; LUMIERE and UCSF locked; UCSF-trained as a reference).

Figure 2. The pathways hold at two external centres without refitting. AUC for detecting the scan at which progression was called, by pathway (clinical data alone, timeline, images, combined) and cohort: MU-Glioma-Post (repeated cross-validation, intended use); LUMIERE and UCSF-ALPTDG (models fitted once on MU-Glioma-Post and applied unchanged); and UCSF-ALPTDG with models trained there, as a reference. At LUMIERE and UCSF-ALPTDG the timeline is the reduced version, and the clinical-only model at LUMIERE is exploratory. At UCSF-ALPTDG the images are shown for the single-scan model and for the image pathway with change features. Bars are 95% patient-bootstrap CIs.

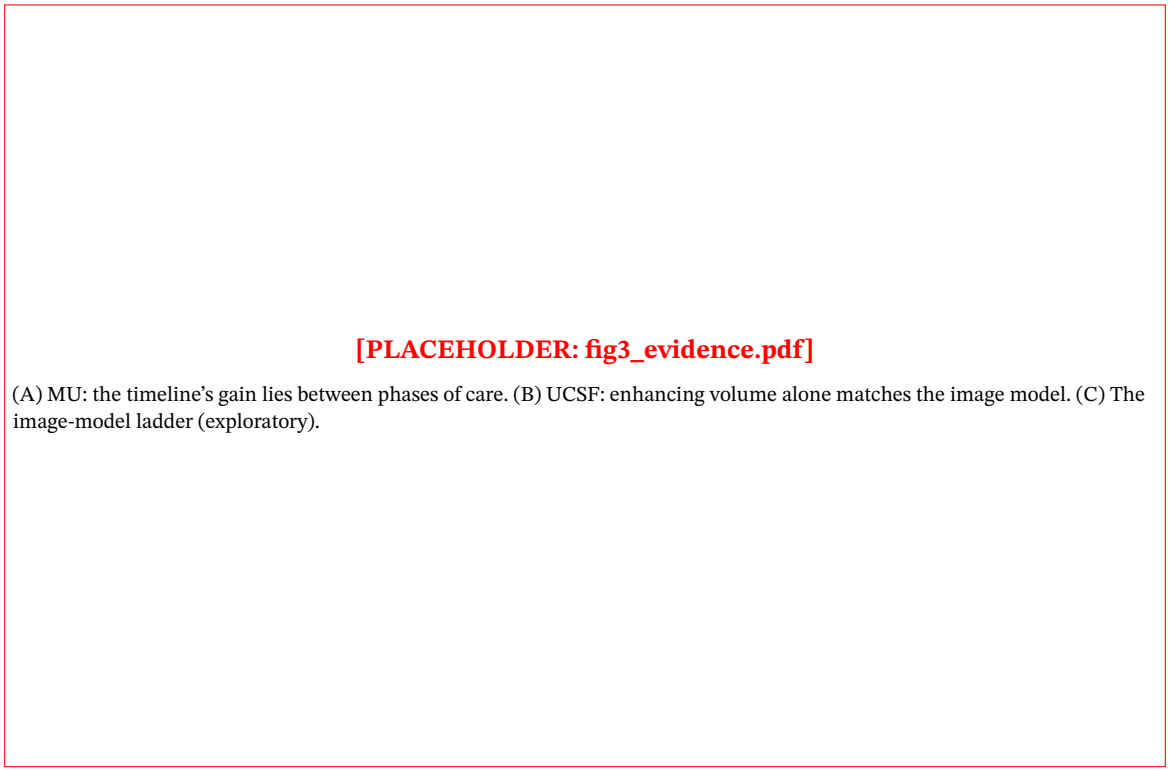


Figure 3. What each score rests on depends on the cohort. (A) MU-Glioma-Post: the gain from adding the timeline to clinical data and images, across phases of care and within them. (B) UCSF-ALPTDG: AUC of enhancing tumour volume alone beside the locked image models, and the share of progression and non-progression scans that UCSF rated as showing increased enhancement. (C) The image model on MU-Glioma-Post built in three steps (current scan; with change from earlier scans; with schedule inputs), across and within phases of care. Panels A and C are exploratory. Bars are 95% patient-bootstrap CIs.

[PLACEHOLDER: fig4_confidence.pdf]

(A) Conformal coverage at the 80% level by cohort and pathway. (B) AUC with and without the disagreement warning. (C) The reliability card: situations by cohort with their evidence tier.

Figure 4. The confidence keeps its coverage and points to doubtful scans. (A) Coverage of the 80% conformal sets by cohort and pathway; the dashed line is the threshold fixed in advance. (B) AUC of the combined score where the pathways agree and where they disagree, and after the disagreeing scans are set aside, compared with removal matched for phase of care. (C) The reliability card: situations (phase of care by change in enhancing tumour) in each cohort, shaded by tier (tested, few cases, not tested). The agreement rule and the card are exploratory.

[PLACEHOLDER: fig5_report.pdf]

One rendered report (segmentation figure, pathways, combined score, confidence) and faithfulness by cohort.

Figure 5. The report states the evidence, faithful to the record. Left: one report as the clinician receives it, with the segmentation figure and its volumes, the two pathway scores, the combined score, the conformal set, the agreement statement and the card's tier. Right: share of reports passing the deterministic check, by cohort, in the evaluation reported in Section 3.5.